REVA University • School of Computing • Department of CSE (AI & ML)

AIML-401: Deep Learning — Revision Study Guide

Topic: **Train Test Split With Code Example**

Candidate ID: STU-807	**Target Exam:** Midterm IA-2 & SEE Exam
Curriculum Unit: Unit 1: Mathematical Foundations	**Compiled:** Nov 02, 2026 - 20:59
Faculty Note Ref: syllabus_overview.pdf	**Platform:** PACD-RAG Academic Intelligence

1. Fundamental Concept & Partitioning Principle

Train-Test Split is a fundamental model evaluation technique that divides an empirical dataset *D* into mutually exclusive partitions: a training set (D_{train}) to fit model parameters and an unseen testing set (D_{test}) to reliably estimate out-of-sample generalization error and prevent overfitting.

2. Python Scikit-Learn Code Implementation

```
# Production Scikit-Learn Implementation
from sklearn.model_selection import train_test_split
import numpy as np

# Partition features X and labels y with fixed seed and stratification
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,        # 80% Training, 20% Testing partition
    random_state=42,      # Ensures deterministic reproducibility
    stratify=y           # Preserves identical class distribution
)
print(f'Train Partition: {X_train.shape}, Test Partition: {X_test.shape}')
```

Key Exam Insight: *Always use stratify=y when handling imbalanced classification datasets to guarantee both train and test splits reflect the true class prior probabilities.*

3. 4-Phase Data Partitioning & Validation Pipeline

Phase	Name	Mathematical / Code Operation	Key Objective
1	Shuffling	Random permutation of index array P = perm(N)	Eliminates chronological or ordered bias
2	Stratification	P(y in D_train) == P(y in D_test) == P(y in D)	Preserves minority class ratios
3	Feature Isolation	D_train intersect D_test == empty set	Prevents target leakage across splits
4	Fit & Evaluate	Scaler.fit(X_train); Scaler.transform(X_test)	Strictly avoids data snooping

4. Mathematical Partition Constraints

Given dataset $D = \{(x_i, y_i)\}_{i=1}^N$, train-test splitting guarantees:

- $D_{train} \cup D_{test} = D$ *and* $D_{train} \cap D_{test} = \emptyset$
- $|D_{test}| / |D| = \theta \in$ *(0.1, 0.3) where standard test ratio* $\theta = 0.20$

5. High-Yield Question & Pitfall Checklist for Exams

Top Exam Trap: Fitting preprocessing transformers (e.g. StandardScaler or PCA) on the entire dataset BEFORE splitting. This causes subtle data leakage resulting in overoptimistic test metrics.

Expected Question (8 Marks): *'Explain the mechanism of train-test split in machine learning. Write Python code using scikit-learn demonstrating stratified splitting, and explain why data leakage occurs if feature normalization is applied before splitting.'*

Marking Rubric:

• 2 Marks: Mathematical justification of generalization vs empirical risk.

• 3 Marks: Valid Python code demonstrating train_test_split with stratify and random_state.

• 3 Marks: Clear explanation of data leakage prevention using Pipeline or split-first paradigm.